

METHODOLOGY

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1. Research Question

Movie rating platforms provide two different perspectives on films: professional critics' evaluations and audience ratings. Critics are generally considered experts, while audience ratings reflect the opinions of a larger and more diverse public. Because disagreements between the two groups are often discussed, we decided to investigate whether critics and audiences actually tend to agree in their evaluation of movies.

Do professional critics agree with movie audiences?

2. Hypotheses

To answer this question, we formulated two hypotheses.

The null hypothesis (H0) states that critics' ratings and audience ratings are independent. In other words, knowing how critics evaluate a movie does not provide information about how audiences evaluate it.

The alternative hypothesis (H1) states that critics' ratings and audience ratings are positively associated. If this hypothesis is true, movies that receive high critics' scores should also tend to receive high audience scores.

3. Dataset

For this project, we used data from Rotten Tomatoes. We worked with two datasets: one containing professional critics' reviews and another containing information about movies, including both Tomatometer scores and audience ratings.

We chose these datasets because they directly provide the two types of evaluations that we wanted to compare. The main variables used in our analysis are the Tomatometer rating, which summarizes critics' evaluations, and the audience rating, which reflects the opinion of viewers. We also used information such as review scores, genres and review dates to better understand the structure of the data.

4. Data Preparation

Before conducting the analysis, we cleaned and prepared the data. First, we merged the critics dataset and the movies dataset in order to gather all relevant information in a single dataframe.

We then removed missing values whenever they prevented the use of a variable. Another important step was the harmonization of critics' scores. Since critics use different rating systems, such as percentages, stars or numerical scales, we created a conversion function that transformed all ratings into a common 0–100 scale. This standardization allowed us to compare reviews consistently across the dataset.

5. Exploratory Analysis

Before testing our hypotheses, we explored the data through descriptive statistics and visualizations.

We first examined the distribution of critics' ratings in order to understand how scores were distributed across the dataset. We also compared Top Critics and regular critics, explored differences between movie genres and observed how the number of reviews evolved over time.

Finally, we created a scatterplot comparing critics' ratings and audience ratings. This graph provided a first visual indication of the relationship between the two variables and suggested a positive association between them.

6. Statistical Analysis

To formally test our hypotheses, we first computed Pearson's correlation coefficient. We chose this measure because both critics' ratings and audience ratings are quantitative variables, and we wanted to evaluate the strength of the linear relationship between them.

After measuring the correlation, we estimated a linear regression model in which audience ratings were treated as the dependent variable and critics' ratings as the explanatory variable. We chose linear regression because it allows us not only to measure the existence of a relationship, but also to evaluate how audience ratings tend to change when critics' ratings increase.

To assess the predictive performance of the model, we divided the dataset into a training set containing 80% of the observations and a testing set containing the remaining 20%. The model was trained on the training set and evaluated on the testing set. This approach allowed us to determine whether the relationship identified by the model generalized well to unseen observations and whether the model suffered from overfitting.

7. Results

The Pearson correlation coefficient between critics' ratings and audience ratings was equal to 0.654, indicating a moderately strong positive relationship. The associated p-value was effectively equal to zero, leading us to reject the null hypothesis of independence.

The linear regression model also revealed a positive relationship between the two variables. The estimated slope coefficient was 0.477, meaning that higher critics' ratings are generally associated with higher audience ratings. The model achieved an R^2 of 0.435 on the training set and an R^2 of 0.396 on the testing set.

Because the training and testing performances are relatively close, there is little evidence of substantial overfitting. The model therefore appears to generalize reasonably well to new observations.

Overall, the results suggest that critics and audiences tend to agree on movie quality. However, the relationship is far from perfect, indicating that audience preferences are influenced by factors other than critics' evaluations alone.